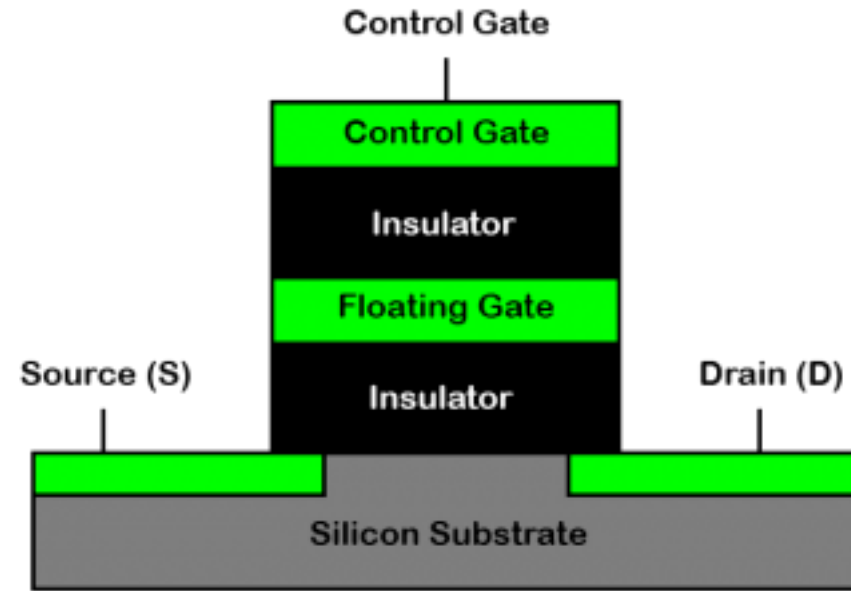


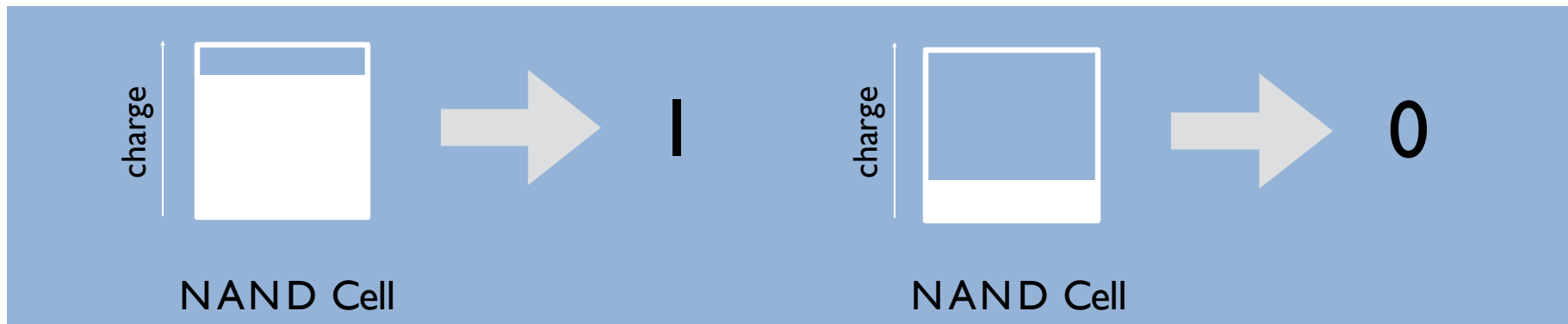
Storage I/O admission using ML

Feb 17 2026

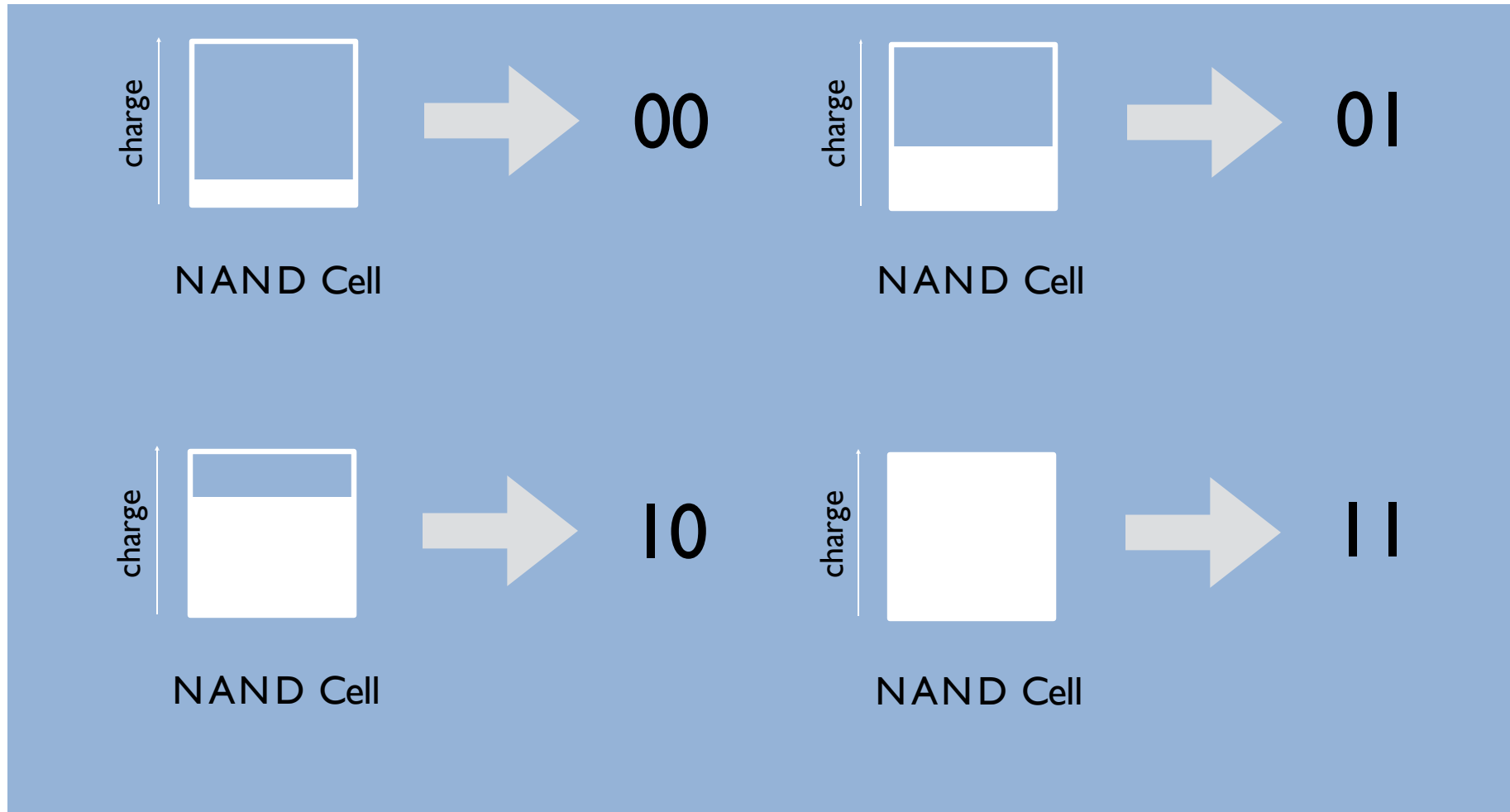
Flash cells



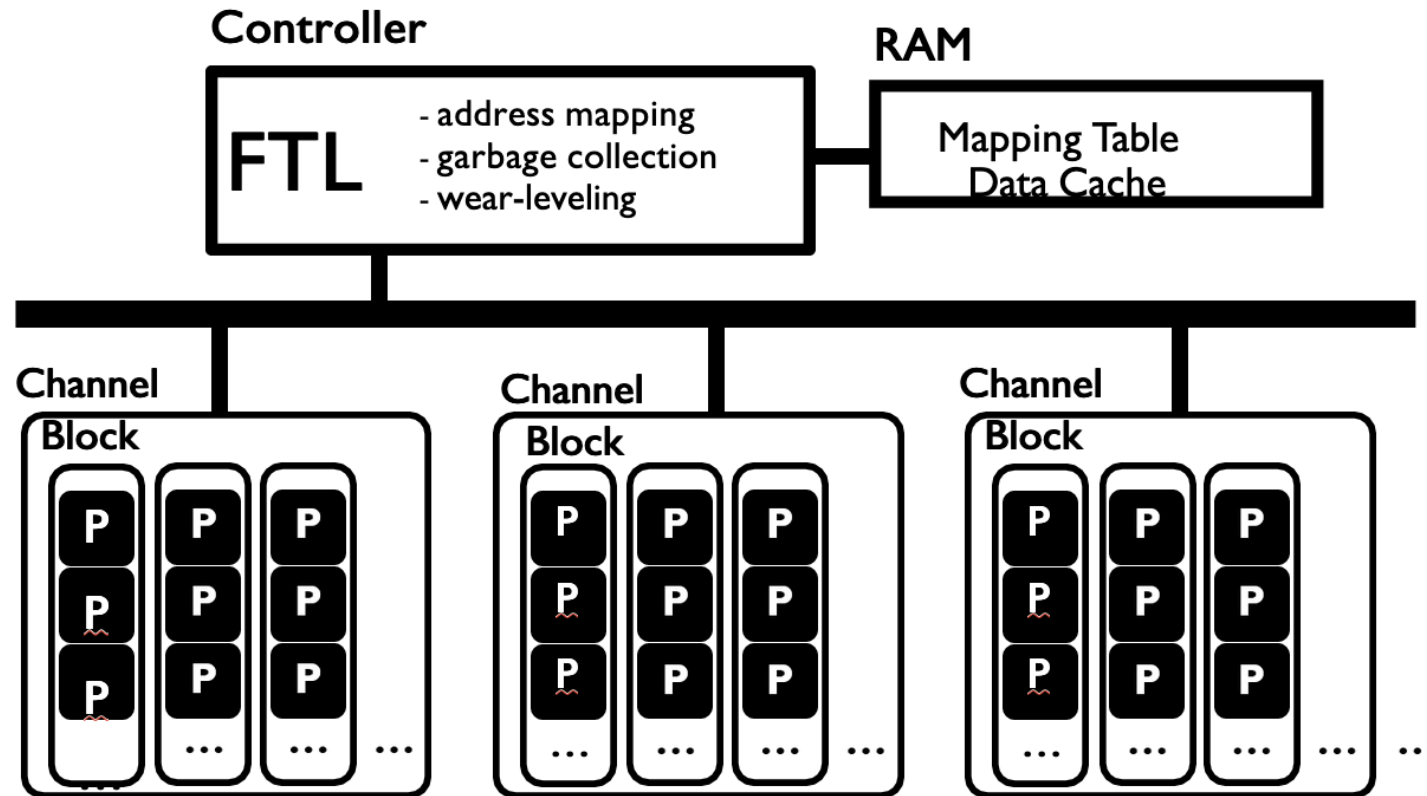
NAND MOSFET
flash cell



Flash cells



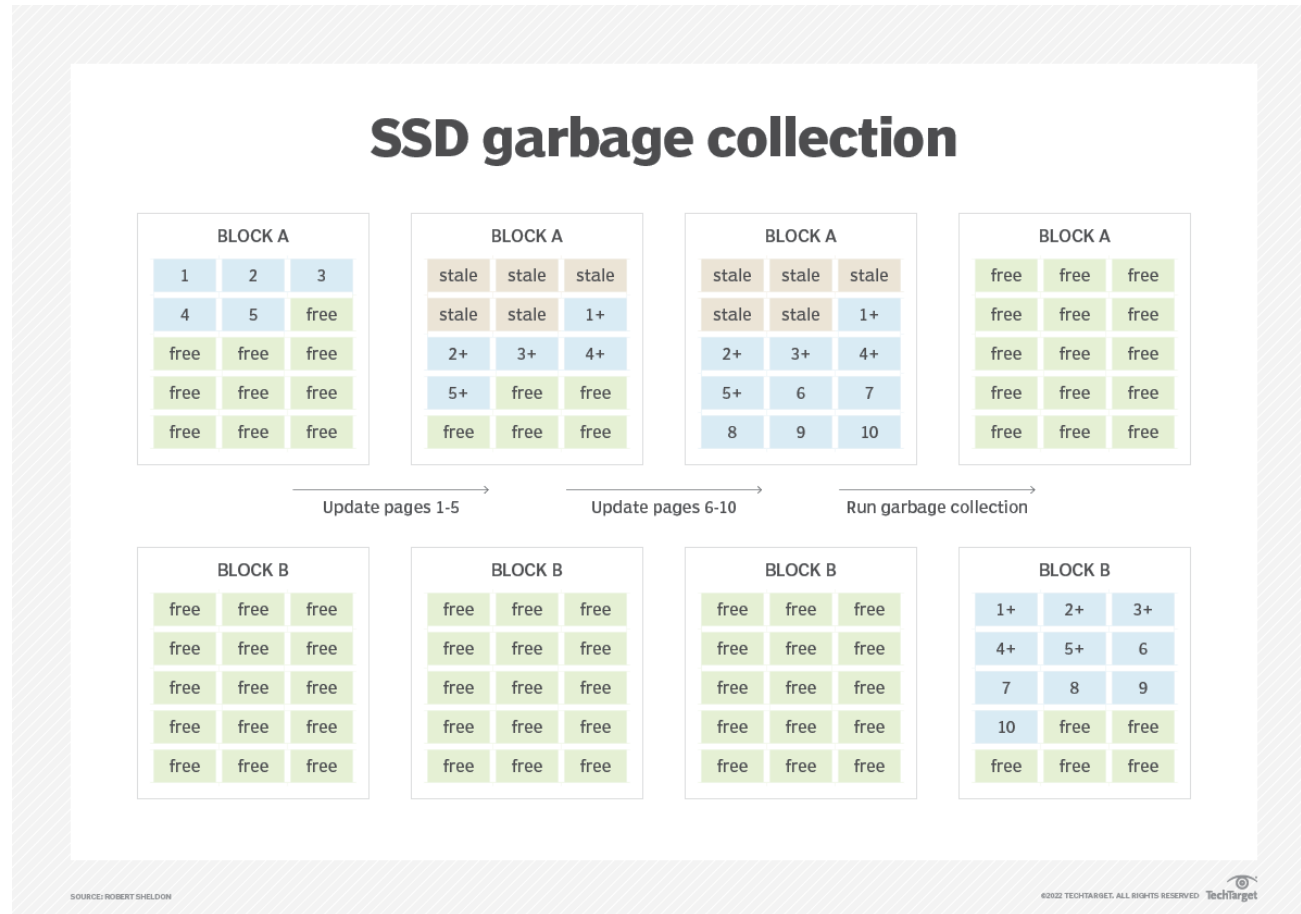
Flash SSD



SSD internal functions

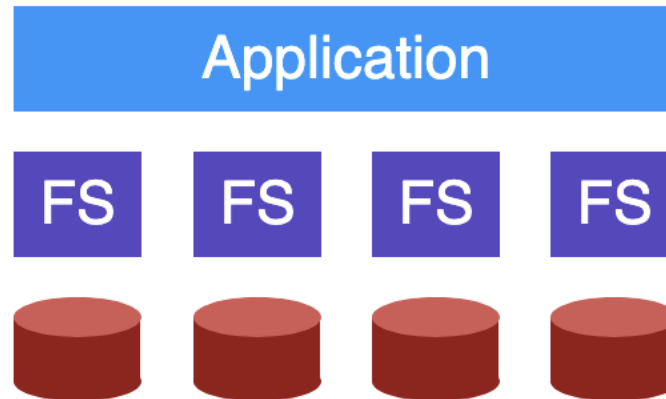
- Buffer-flushing
- Wear-levelling
 - Flash cells have a limited write endurance (~10K erases)
- Garbage collection
 - To avoid repeated writes to same cell, data is copied to new pages
 - Old pages are considered *invalid*
 - When space low, "free" invalid pages -> erase the entire block -> copy valid pages to another block
 - When to run?
 - Too frequent -> frequent erases -> early wear-out
 - Infrequent -> low space -> I/O latency increases

Garbage collection



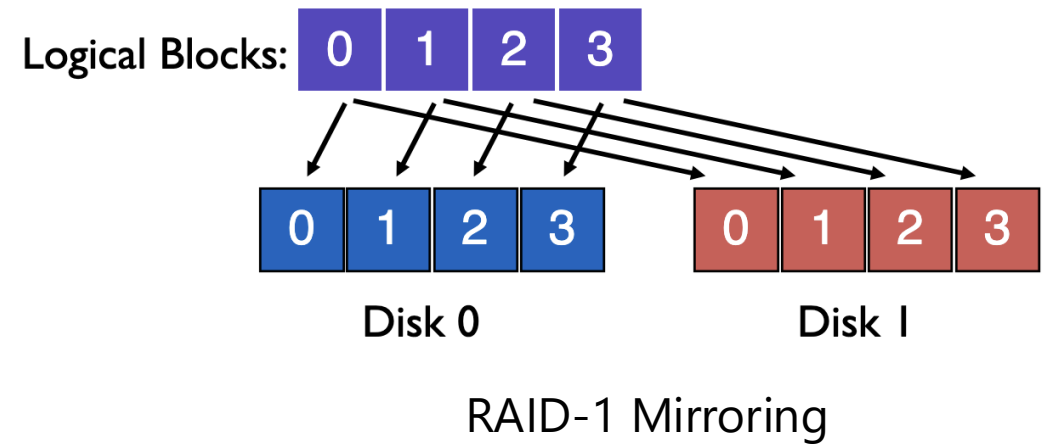
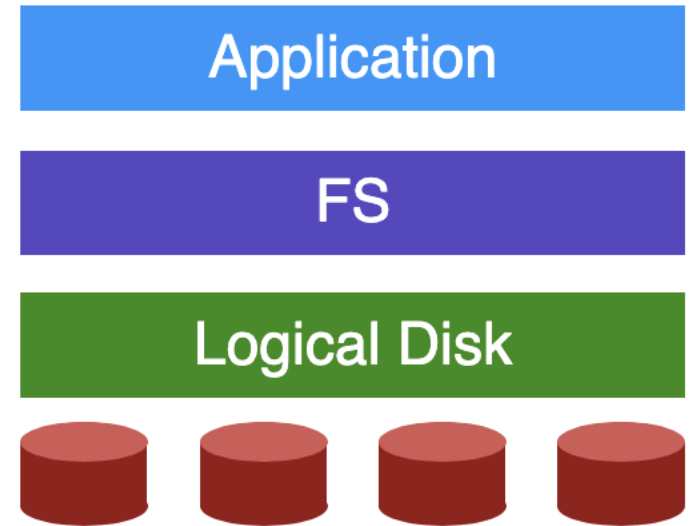
RAID

- Need multiple disks for:
 - Higher capacity, reliability, performance
- Option 1: Independent devices and FSes



RAID

- Logical disk backed by multiple physical disks
 - Less burden on application
- RAID-0: Striping
- RAID-1: Mirroring
- RAID-4: Parity
- RAID-5: Parity rotation



Admission control: LinnOS

- Goal: Predict I/O latency based on last few I/Os
- Approach: Use a neural net
 - Features: pending I/Os, latency of recent I/Os
 - Prediction: is I/O latency greater than *inflection point*?

Discussion

- For long-running workloads, Heimdall uses periodic re-training. What are some ways to improve this?
- How to systematically explore the different stages in ML pipelines?